ALKENE

21. (b) $$CH_2=CH_2+\underset{\text{alk.}}{KMnO_4}\xrightarrow[H_2O]{[O]}\underset{\underset{OH}{|}}{CH_2}-\underset{\underset{OH}{|}}{CH_2}+\underset{\text{Brown}}{MnO_2}$$

Colour less

$$CH\equiv CH+\underset{alk.}{KMnO_4}\xrightarrow{[O]}\underset{\underset{COOH}{|}}{COOH}+\underset{\text{Brown}}{MnO_2}$$

Oxalic acid

22. (d) Octane number increases in the order

Straight chain alkanes < Branched chain alkanes < Olefins < Cyclo alkanes < Aromatic compounds

Since, straight chain alkane has minimum octane number. Hence, it produces maximum knocking.

23. (c) $$R-CH=CH-R\xrightarrow[\text{room temp.}]{\text{dil. aqueous } KMnO_4}R-\underset{\underset{OH}{|}}{CH}-\underset{\underset{OH}{|}}{CH}-R$$

(Alcohol)

$$R-CH=CH-R\xrightarrow[\text{heat}]{\text{Conc. } KMnO_4}R-COOH+R-COOH$$

24. (b) $$\underset{\text{2-methyl-1-butane}}{CH_2=\overset{\overset{CH_3}{|}}{C}-CH_2-CH_3}+H_2O\xrightarrow[\text{Markowniko ff's rule}]{H_2SO_4}CH_3-\overset{\overset{CH_3}{|}}{\underset{\underset{OH}{|}}{C}}-CH_2-CH_3$$

2-Methyl-2-butanol

25. (c) By dehydration with conc. H_2SO_4 at 170°C

Ethanol loses water to form ethene.

26. (a) Detection of double bonds

Dilute Potassium permanganate is used to detect unsaturation.

27. (b) $CH_3-CH=CH_2+H_2O \xrightarrow[\text{Markowniko ff's rule}]{\text{Conc. } H_2SO_4} \underset{\text{Isopropyl alcohol}}{CH_3-\underset{\underset{OH}{|}}{CH}-CH_3}$ 28

29 (b) Trans-2-butene

Most stable alkene gives least heat.

30 (a) Iodine in CCl_4

Does not react with alkene or alkane, so no distinction.

31. (c) $CH_3–CH_2–CH=CH–CH_2–CH_3$

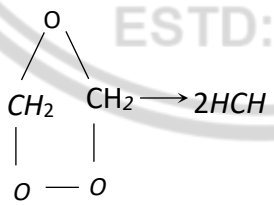
Symmetrical alkene gives same carboxylic acid on oxidation.

32. (d) $CH_3-\underset{\underset{Br}{|}}{CH}-CH_2-CH_3 \xrightarrow{alc.KOH} \underset{\text{2-butene}}{CH_3CH=CHCH_3}$

The reaction is dehydrohalogenation.

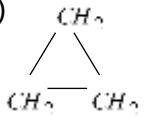
33. (c) Cyclohexene

1,3-butadiene reacts with ethene via Diels–Alder reaction to form a six-membered ring (cyclohexene).

34. (a) $CH_2=CH_2+O_3 \longrightarrow$ 

35. (c) Oil are unsaturated esters which are converted into fats by saturating it by catalytic hydrogenation.

36. (c) $>C=C< \xrightarrow[H]{\text{Hydrogenation}} >C-C<$

37. (a) 

Cyclo Propane

38. (b) *e.g.* $CH_2 = CH_2 + Br_2 \rightarrow \underset{Br}{\underset{|}{CH_2}} - \underset{Br}{\underset{|}{CH_2}}$

Half of the double bond is broken. It means π bond is broken while sigma bond is retained also two new $C - Br$ bonds are formed.

39. (c) CH_3–CDCl–CH$(CH_3)_2$

Addition of DCl follows Markovnikov's rule.

D attaches to the carbon with more hydrogens, and Cl goes to the more substituted carbon, forming the given product.

40. (b) $CH_3 - \overset{Br}{\overset{|}{CH}} - CH_2 - CH_3 + \underset{(alc)}{KOH} \rightarrow \underset{\text{Butene-2}}{CH_3 - CH = CH - CH_3} + KBr + H_2O$